

OOPS(PBL) PROGRAM - 4

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Topic: Write a program in C++ to perform following operations on complex numbers Add, Subtract, Multiply, Divide, Complex conjugate. Design the class for complex number representation and the operations to be performed.

INPUT :

```
#include <iostream>
using namespace std;
class Complex {
private:
    double real;
    double imag;
public:
    Complex(double r = 0.0, double i =
0.0) : real(r), imag(i) {}
    void read() {
        cout << "Enter real part: ";
        cin >> real;
        cout << "Enter imaginary part: ";
        cin >> imag;
    }
    void display() const {
        if (imag >= 0)
            cout << real << " + " << imag
<< "i\n";
        else
            cout << real << " - " << -imag
<< "i\n";
    }
    Complex add(const Complex &c)
const {
```

```
        return Complex(real + c.real,
imag + c.imag);
    }
    Complex subtract(const Complex
&c) const {
        return Complex(real - c.real,
imag - c.imag);
    }
    Complex multiply(const Complex
&c) const {
        return Complex(real * c.real -
imag * c.imag,
            real * c.imag + imag *
c.real);
    }
    Complex divide(const Complex
&c) const {
        double denom = c.real * c.real +
c.imag * c.imag;
        if (denom == 0) {
            cout << "Division by zero
error!\n";
            return Complex(0, 0);
        }
    }
```

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        return Complex((real * c.real +
imag * c.imag) / denom,
        (imag * c.real - real *
c.imag) / denom);
    }
    Complex conjugate() const {
        return Complex(real, -imag);
    }
};

int main() {
    Complex c1, c2;
    cout << "Enter First Complex
Number:\n";
    c1.read();
    cout << "Enter Second Complex
Number:\n";
    c2.read();
    cout << "\nFirst: ";
    c1.display();
    cout << "Second: ";
    c2.display();
    cout << "\nAddition: ";
    c1.add(c2).display();
    cout << "Subtraction: ";
    c1.subtract(c2).display();
    cout << "Multiplication: ";
    c1.multiply(c2).display();
    cout << "Division: ";

```

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c1.divide(c2).display();
cout << "Conjugate of First: ";
c1.conjugate().display();
return 0;
}

```

OUTPUT :

Output

```

Enter First Complex Number:
Enter real part: 12
Enter imaginary part: 33
Enter Second Complex Number:
Enter real part: 12
Enter imaginary part: 33

```

```

:: First: 12 + 33i
Second: 12 + 33i

```

```

Addition: 24 + 66i
Subtraction: 0 + 0i
Multiplication: -945 + 792i
Division: 1 + 0i
Conjugate of First: 12 - 33i

```